INCH-POUND

MIL-S-70587B (AR)
27 July 1991
SUPERSEDING
See 6.6

MILITARY SPECIFICATION

SERVO AMPLIFIER: 9361156

This specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the servo amplifier used on Gun, Anti-Aircraft, Artillery, Self-Propelled 20MM, M163A2, and Towed 20MM, M167A2. The servo amplifier circuits provide the final amplification of the drive signal for the servo motor. The output signal shall be proportional to the drive signal input from the control circuits under constant load. The output signal is polarized to reflect the direction of the input signal and drive the servo motor in the corresponding direction (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-13855 - Weapons: Small Arms and Aircraft Armament Subsystems, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization, Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9360958 - Electrical Servo Amplifier Schematic
9361156 - Servo Amplifier Assembly
12000117 - Integrated Circuit, Operational Amplifier Instrumentation

SPECIAL PACKAGING INSTRUCTIONS

PS9361156 - Packaging of Servo Amplifier: 9361156

(Copies of drawings and packaging data sheets required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.



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3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9361156, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the servo amplifier shall meet the requirements of MIL-W-13855 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The servo amplifier shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

| Frequency Range | Amplitude |
|---|---|
| 10 to 15 hertz | 0.06 inches double amplitude |
| 15 to 25 hertz | 0.04 inches double amplitude |
| 25 to 55 hertz | 0.02 inches double amplitude |

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 10-55-10 hertz sweep time is 3 minutes 45 seconds ± 10%.

3.3.2 Shock. The servo amplifier shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The servo amplifier shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-operating. The servo amplifier shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The servo amplifier shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 deg.F and +155 deg.F.



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3.3.4.2 Non-operating (storage). The servo amplifier shall not suffer permanent damage when subjected to the constant temperature extremes of -60 deg.F for a period of 24 hours and +160 deg.F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The servo amplifier shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram (9360958) of the servo amplifier.

3.5 Performance.

3.5.1 Functional description. The servo amplifier shall contain the circuits necessary to process signals generated by a hand control potentiometer in the control assembly and supply either a clockwise (CW) or counterclockwise (CCW) output, which is proportional to the input voltage difference, to the field windings of the drive motor. These circuits shall also monitor and sustain signal level voltages, and provide overcurrent protection for the servo amplifier system and the drive motor. Refer to Table I (Connector J1 pin designations) for connection and pin designations.

TABLE I. Connector J1 pin designations.

| PIN | FUNCTION |
|---|---|
| B | Motor voltage sense point (low) |
| C | Gun command (low) |
| E | Motor voltage sense point (hi) |
| H | Gun command (hi) |
| J | Drive motor return |
| K | Drive motor (hi) |
| L | Drive motor (low) |
| M | Servo amplifier power (+28 Vdc) |
| V | Current feedback power |



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3.5.2 Performance characteristics.

3.5.2.1 Signal Component Assembly, A1. The servo amplifier shall contain the amplifier printed circuit board assembly (A1) to provide the following:

a. Amplifier circuits to increase signal levels.
b. A feedback network to provide motor current proportional to input voltage.
c. Current limiting functions.

3.5.2.1.1 Linear amplifiers. The A1 board shall contain linear amplifiers to increase signal level voltages.

3.5.2.1.2 Input differential amplifier. The A1 board shall contain an input differential amplifier to amplify the input command signal. It shall have a gain of 1.9 ± 1%.

3.5.2.1.3 Summing amplifier. The A1 board shall contain a summing amplifier to produce the difference between the output signal of the differential amplifier and a current feedback signal. It shall have a gain of 1.04 ± 2%.

3.5.2.1.4 Lag-lead amplifier. The A1 board shall contain a lag-lead amplifier to provide gain and frequency response shaping, necessary to close the current feedback loop in a stable manner. It shall have a gain of 15 ± 2%, a lag frequency of 1.6 Hz ± 11% and a lead frequency of 81.5 Hz ± 11%.

3.5.2.1.5 Feedback differential amplifier. The feedback network of the A1 board shall contain differential amplifier circuits to increase the level of the resultant feedback differential signal in the presence of +28 Vdc common mode voltage. It shall have a gain of 0.25 ± 2%.

3.5.2.1.6 Feedback amplifier. The feedback network of the A1 board shall contain a feedback amplifier to supply the necessary signal level to the input and output sides of the summing amplifier. It shall have a gain of 133 ± 2%.

3.5.2.1.7 Motor voltage sensors. The feedback network of the A1 board shall contain diode/capacitor networks to provide an indication of applied motor voltage between power pulses. They shall provide a dc output proportional to the power pulse widths minus 0.65 ± 10% volts diode drop.

3.5.2.1.8 Current limiting functions. The A1 board shall contain current limiting circuits to provide the following:

a. Overcurrent protection for the servo amplifier system.
b. Prevent the current supplied to the drive motor from exceeding 135 amperes.
c. Regulate signal level voltages.



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3.5.2.2 Power Supply and Driver Component Assembly, A2 The servo amplifier shall contain the power supply/driver circuit board assembly (A2) to provide the following;

a. ±15 Vdc for all servo amplifier signal level circuits.
b. CW and CCW power pulses.
c. Current limiting functions.

3.5.2.2.1 Pre-regulator. The A2 board shall contain a pre-regulator to regulate the input voltage of +28 Vdc down to +19.3 Vdc to supply the dc-dc converter and lockout delay circuits.

3.5.2.2.2 Dc-dc converter. The A2 board shall contain a dc-dc converter to supply ±18 Vdc to the dual regulator with an input of 20.0 Vdc ± 3.0 Vdc.

3.5.2.2.3 Dual regulator. The A2 board shall contain a dual regulator to supply ±15 Vdc ± 1.3% needed for all signal level circuits.

3.5.2.2.4 Triangle wave generator. The A2 board shall contain a triangle wave oscillator to produce a 2.5 to 2.8 V p-p, with negative peak 0.1 to 0.4 above ground, 100±625Hz triangle wave for input to the pulse width modulator.

3.5.2.2.5 Pulse width modulator. The A2 board shall contain a pulse width modulator to convert the signal from the lag-lead amplifier to CW or CCW pulses that have a duty cycle proportional to the voltage input as specified below.

| Input | Duty Cycle Outputs |
|---|---|
| 2.50 ± 0.03 Vdc | 100 - 10 |
| 1.71 ± 0.02 Vdc | 65 ± 10 |
| 0.80 ± 0.01 Vdc | 30 ± 10 |
| 0.00 ± 0.01 Vdc | 0 + 2 |

3.5.2.2.6 Drive stage. The A2 board shall contain CW and CCW drive stages to increase the power level of the positive and negative output pulses from the pulse width modulator.

3.5.2.2.7 Lockout functions. The A2 board shall contain circuits to provide the 90 msec delay of turret motion each time power is applied to the servo amplifier to allow electronic circuits in the system to stabilize.



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3.5.2.3 Interconnection printed circuit board assembly. The servo amplifier shall contain the interconnection printed circuit board (A3) to provide interfacing for the servo amplifier circuitry.

3.5.2.4 Power blocks. The servo amplifier shall contain CW and CCW power blocks to increase the current to the level required to operate the turret drive motor up to 250A.

3.5.3 Test points. The servo amplifier shall contain test points as indicated on schematic 9360958.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task)F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-W-13855 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.



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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

| | | |
|---|---|---|
| a. | Temperature | +50 deg. to +90 deg.F |
| b. | Relative humidity | 90% RH max. |
| c. | Pressure | 28 to 32 inches of Hg |

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) servo amplifiers for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table I (Functional Test) and Table II (Special Sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any of the stated requirements.



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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specification and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-W-13855 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table II shall be performed on a 100% basis.

TABLE II. Functional tests (100 percent inspection).

| CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|
| Critical: | None | N/A |
| Major: | | |
| 101. Electrical | 3.4 | 4.7.2 |
| 102. Performance | 3.5 | 4.7.3 |
| Minor: | None | N/A |



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4.5 Special sampling. One servo amplifier out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table II and shall then meet the tests in Table III. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE III. Special sampling.

| CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|
| Critical: | None | N/A |
| Major: | | |
| 103. Vibration | 3.3.1 | 4.7.1.1 |
| 104. Shock | 3.3.2 | 4.7.1.2 |
| 105. Altitude | 3.3.3 | 4.7.1.3 |
| 106. Temperature | 3.3.4 | 4.7.1.4 |
| 107. Humidity | 3.3.5 | 4.7.1.5 |
| Minor: | None | N/A |

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-W-13855 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in



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the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table IV (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE IV. Test equipment list.

| ITEM NUMBER | DESCRIPTION |
|---|---|
| 1. Digital voltmeter | Input impedance: 10 megohm minimum. Range: 1 mV to 28 V. |
| 2. Oscilloscope | Input impedance: 1 megohm minimum, dual time base with mixed sweep. Capability range: 0 to 30 Vdc, 20 MHz band width minimum. |
| 3. Power Supplies (2) | ±15 Vdc at 1 Adc, Regulated 0.01 Vdc<br>28 Vdc at 40 Adc, Regulated 0.01 Vdc |

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The servo amplifier shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the servo amplifier shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.



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4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the servo amplifier shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. The servo amplifier performance shall be tested in accordance with the following using the test equipment listed in 4.6.2.3.



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4.7.3.1 Signal Component Assembly (A1). The A1 board shall be tested as follows in accordance with the requirements of 3.5.2.1.
NOTE: All measurements are with respect to TP6.

a. Connect the A1 board to the fixture described by Figure 1, switch S2 to A, switch S3 to A, switch S6 to B, switch S7 to B, and switch S1 to ON.

b. Apply 0.000 ± 0.001 Vdc to TP10. Apply a variable DC voltage to TP9 and adjust it until the voltage at TP1 is 0.00 ± 0.001 Vdc. Verify that the voltage at TP9 is 0.00 ± 0.01 Vdc.

c. Set switch S6 to A, switch S7 to A, and switch S4 to position 1. Adjust R23 until the voltage at TP7 is 1.00 ± 0.001 Vdc. Adjust R12 until the voltage at TP5 is 0.000±0.001 Vdc.

d. The voltage at TP1 shall exhibit a dc gain characteristic of 30 ± 2 when tested in accordance with Table V.

e. Set switch S2 to B, switch S3 to B, switch S4 to position 2, and switch S5 to position 3. The voltage at TP1 shall be 0.00 Vdc maximum (negative).

f. Set S4 to position 3. The voltage at TP3 shall be 4.8 ± 1.0 Vdc.

g. Set switch S2 to A, switch S3 to A, switch S4 to position 2, and switch S5 to position 3. The voltage at TP1 shall be 0.00 Vdc maximum (positive).

h. Set S4 to position 3. The voltage at TP2 shall be 4.8 ± 1.0 Vdc.

i. The input waveform at TP8 shall be a square wave of: peak amplitude 9 V minimum, frequency 1 Hz ± 20 percent, rise time 10 ms maximum.

j. Set switch S4 to position 1, Switch S7 to B and S5 to position 5. The leading edge of the waveform at TP1 shall display a rise time of 150 msec minimum (10 percent and 90 percent points).



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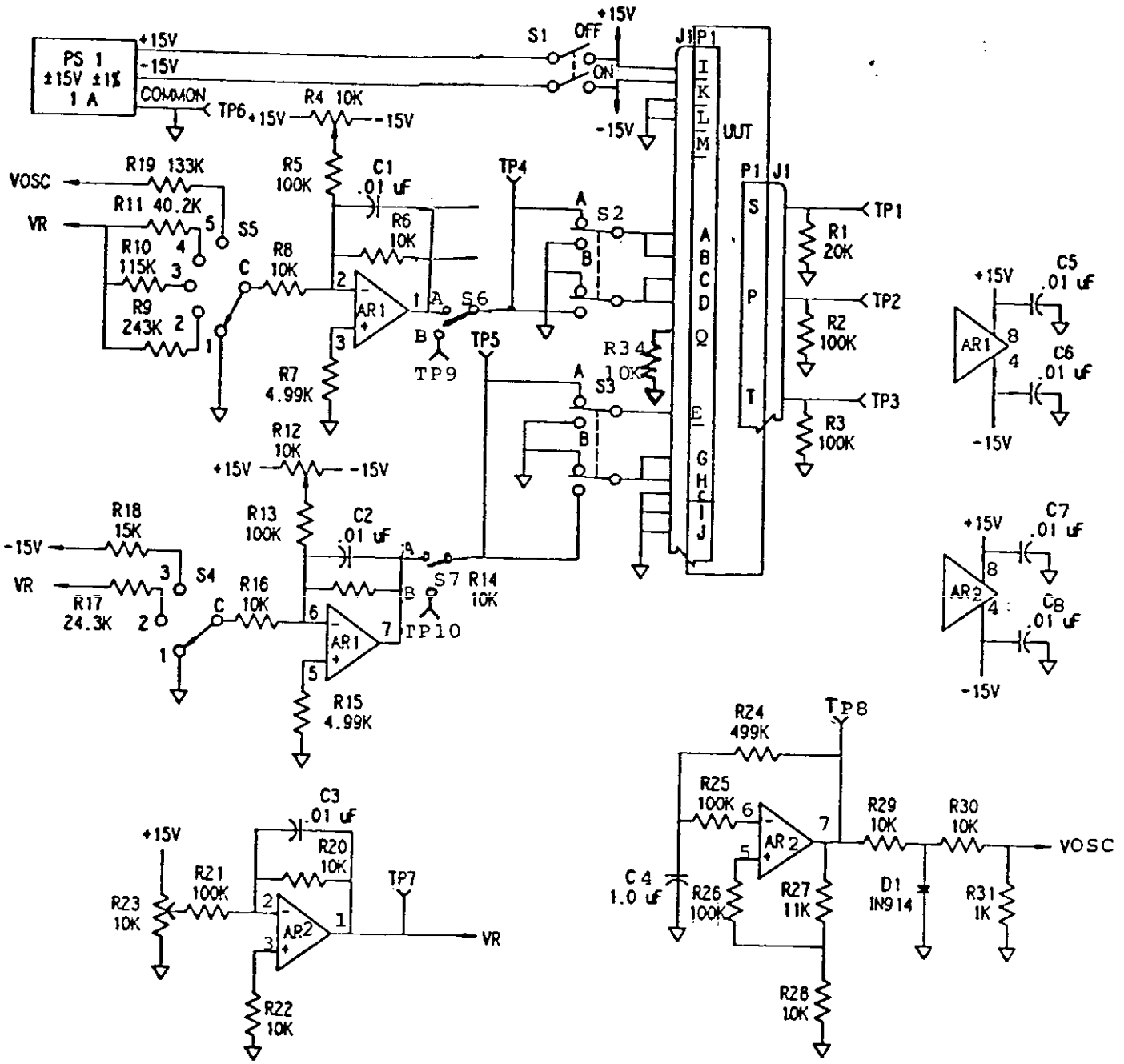



FIGURE 1. A1 test setup (sheet 1 of 2.



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| | | | |
|---|---|---|---|
| R1 | 20 Kohms | R24 | 499 Kohms |
| R2 | 100 Kohms | R25 | 100 Kohms |
| R3 | 100 Kohms | R26 | 100 Kohms |
| R4 | 10 Kohms, 20-turn trimmer | R27 | 11 Kohms |
| R5 | 100 Kohms | R28 | 10 Kohms |
| R6 | 10 Kohms | R29 | 10 Kohms |
| R7 | 4.99 Kohms | R30 | 10 Kohms |
| R8 | 10 Kohms | R31 | 1 Kohms |
| | | R34 | 10 Kohms |
| R9 | 243 Kohms | C1 | .01 uF |
| R10 | 115 Kohms | C2 | .01 uF |
| R11 | 40.2 Kohms | C3 | .01 uF |
| R12 | 10 Kohms, 20-turn trimmer | C4 | 1.0 uF |
| R13 | 100 Kohms | C5 | .01 uF |
| R14 | 10 Kohms | C6 | .01 uF |
| | | C7 | .01 MF |
| | | C8 | .01 MF |
| R15 | 4.99 Kohms | AR1 | 12000117-3 or equivalent |
| | | AR2 | 12000117-3 or equivalent |
| R16 | 10 Kohms | D1 | 1N914 |
| R17 | 24.3 Kohms | S1 | DPDT |
| R18 | 15 Kohms | S2 | DPDT |
| R19 | 133 Kohms | S3 | DPDT |
| R20 | 10 Kohms | S4 | SP3T |
| R21 | 100 Kohms | S5 | SP5T |
| R22 | 10 Kohms | S6 | SPDT |
| | | S7 | SPDT |
| R23 | 10 Kohms, 20-turn trimmer | J1 | M55302/05 |

Note: All fixed resistors ±1%, 1/4 watt.
Note: All capacitors 50 V ceramic ±10%.

FIGURE 1. A1 test setup (sheet 2 of 2).



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TABLE V. DC Gain

| S2 POSITION | S5 POSITION | TP4 mVdc | TP1 Vdc |
|---|---|---|---|
| A | 1 | 0±0.01 | 0.0 ± 0.3 |
| A | 2 | 39.5±2% | -1.187 ± 0.103 |
| A | 3 | 80.0±2% | -2.403 ± 0.208 |
| A | 4 | 199.0±2% | -4.20 max. (saturation) |
| B | 1 | 0 ±0.01 | 0.0 ± 0.3 |
| B | 2 | 39.5±2% | 1.187 ± 0.103 |
| B | 3 | 80.0±2% | 2.403 ± 0.208 |
| B | 4 | 199.0±2% | 5.0 maximum (saturation) |

NOTE: Adjust R4 to obtain required values.



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4.7.3.2 Power supply and driver component assembly (A2). The A2 board shall be tested as follows in accordance with the requirements of 3.5.2.2.

a. Connect the A2 board to the fixture described by Figure 2, switch S2 to position 4, and switch S1 to ON.

b. Voltage at terminal O with respect to terminal K shall be 20.0 ± 3.0 Vdc.

c. Voltage at terminal G with respect to terminal K shall be 15.00 ± 0.20 Vdc, and voltage at terminal I with respect to terminal K shall be -15.00 ± 0.20 Vdc.

d. Voltage measured in step c shall track within 0.180 Vdc.

e. Waveform at terminals U, V shall have a frequency of 625 ± 100 Hz an amplitude of 2.5 to 2.8 Vdc with a negative peak 0.1 to 0.4 Vdc.

f. Terminals M, Q shall exhibit a 90 ± 30 msec delay (50% points) with respect to terminal O when S1 is switched from OFF to ON.

g. The voltage at terminals W, X and terminals Y, Z with respect to terminal K shall exhibit the duty cycle characteristics specified by Table VI (Duty cycle - terminals W, X, Y, Z), high level voltage shall be 27.0 volts minimum and the low level voltage shall be 0.10 volts maximum. To preclude thermal damage, operate units for 5-10 seconds when at high duty cycles.

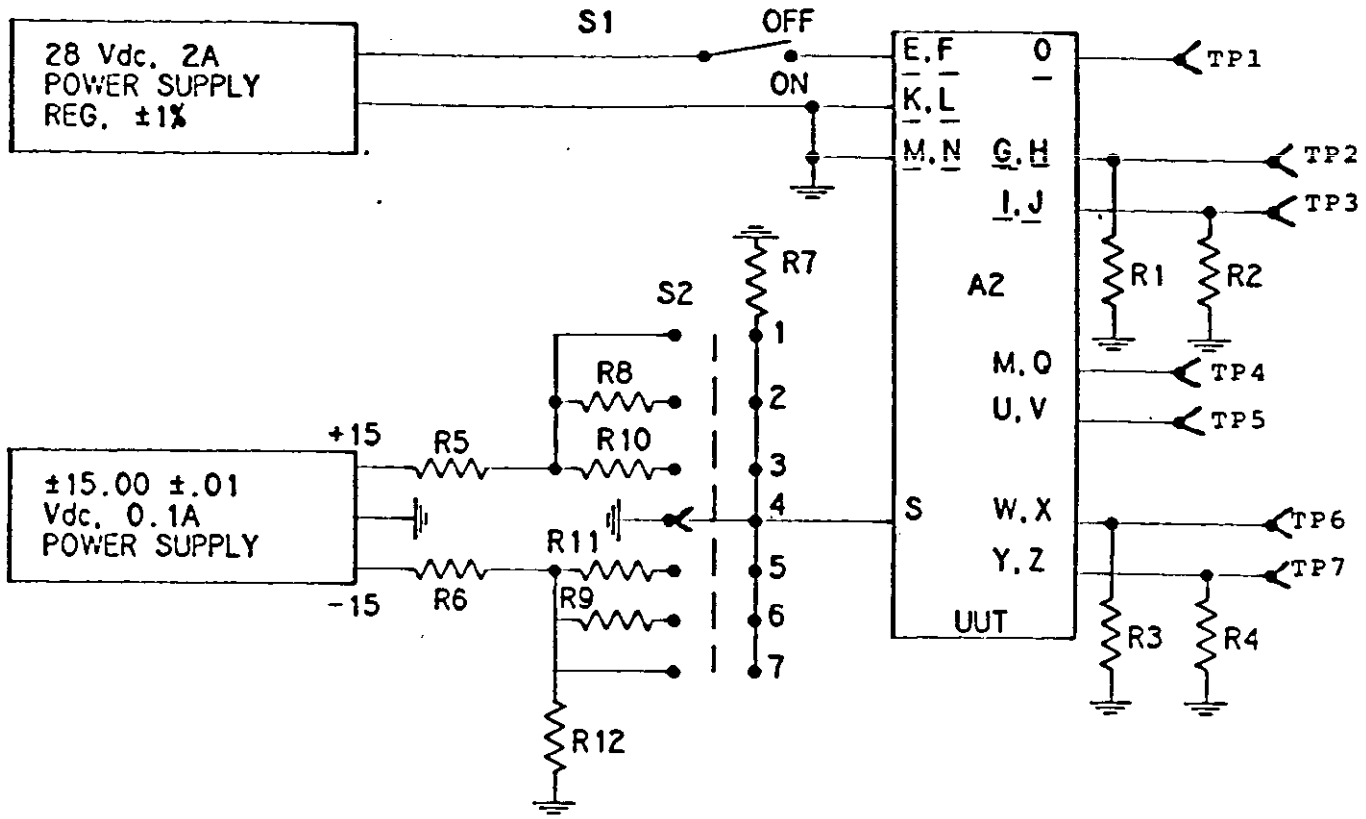
TABLE VI. Duty cycle - terminals W, X, Y, Z.

| S2 POSITION | S | W, X | Y, Z |
|---|---|---|---|
| 1 | 2.50 ± 0.03 Vdc | 100 - 10 | 0 |
| 2 | 1.71 ± 0.02 Vdc | 65 ± 10 | 0 |
| 3 | 0.80 ± 0.01 Vdc | 30 ± 10 | 0 |
| 4 | 0.00 ± 0.01 Vdc | 0 + 2 | 0 + 2 |
| 5 | -0.80 ± 0.01 Vdc | 0 | 30 ± 10 |
| 6 | -1.71 ± 0.02 Vdc | 0 | 65 ± 10 |
| 7 | -2.50 ± 0.03 Vdc | 0 | 100 - 10 |

NOTE: Duty cycle refers to the ratio of time-on to the period expressed in percent.



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NOTE: Adjust R4 to obtain required values.

| | | | |
|---|---|---|---|
| R1, R2 | 300 ohms ± 5%<br>2 watt | R8, R9 | 2.80 Kohms ± 1%<br>1/2 watt |
| R3, R4 | 30 ohms ± 5%<br>50 watt | R10, R11 | 12.7 Kohms ± 1%<br>1/2 watt |
| | | R12 | 20.0 Kohms ± 1%<br>1/2 watt |
| R5, R6 | 4990 ohms ± 1%<br>1/2 watt | S1 | SPDT Toggle |
| | | S2 | SP7T (Shorting) |
| R7 | 1050 ohms ± 1%<br>1/2 watt | | |

Note: Use connector M55302/6-05 to interface unit under test (UUT) to fixture.

FIGURE 2. A2 test setup.



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4.7.3.3 Interconnection printed circuit board (A3). The A3 board shall be tested for continuity as specified by the schematic diagram, 9360958, in accordance with the requirements of 3.5.2.3.

4.7.3.4 Servo amplifier. The servo amplifier shall be tested as follows in accordance with the requirements of 3.5.2.4. NOTE: All measurements are with respect to TP7.

a. Connect the servo amplifier to the fixture described by Figure 3. Set switch S4 to position 1, switch S3 to A, switch S1 to ON, and switch S2 to ON.

b. Adjust R6 until TP6 equals 0.00 ± 0.01 Vdc.

c. The outputs shall exhibit the duty cycle characteristics specified by Table VII (Duty cycle - terminals K, L, V, E, B).

TABLE VII. Duty cycle - terminals K, L, V, E, B.

| S4 POSITION | S3 POSITION | TP4 DUTY CYCLE % (K) | TP5 DUTY CYCLE % (L) | TP1 VDC (V) | TP2 VDC (E) | TP3 VDC (B) |
|---|---|---|---|---|---|---|
| 1 | A | 0 + 2 | 0 + 2 | 0.0 ± 0.1 | 0.0 + 2.0 | 0.0 + 2.0 |
| 2 | A | 0 | 35 ± 10 | 1.0 ± 0.1 | 0.0 + 2.0 | 8.0 ± 1.6 |
| 3 | A | 0 | 65 ± 10 | 2.0 ± 0.2 | 0.0 + 2.0 | 16.0 ± 3.2 |
| 4 | A | 0 | 100 - 10 | 2.7 max. | 0.0 + 2.0 | 25.0 ± 5.0 |
| 1 | B | 0 + 2 | 0 + 2 | 0.0 ± 0.1 | 0.0 + 2.0 | 0.0 + 2.0 |
| 2 | B | 35 ± 10 | 0 | -1.0 ± 0.1 | 8.0 ± 1.6 | 0.0 + 2.0 |
| 3 | B | 65 ± 10 | 0 | -2.0 ± 0.2 | 16.0 ± 3.2 | 0.0 + 2.0 |
| 4 | B | 100 - 10 | 0 | -2.7 min. | 25.0 ± 5.0 | 0.0 + 2.0 |

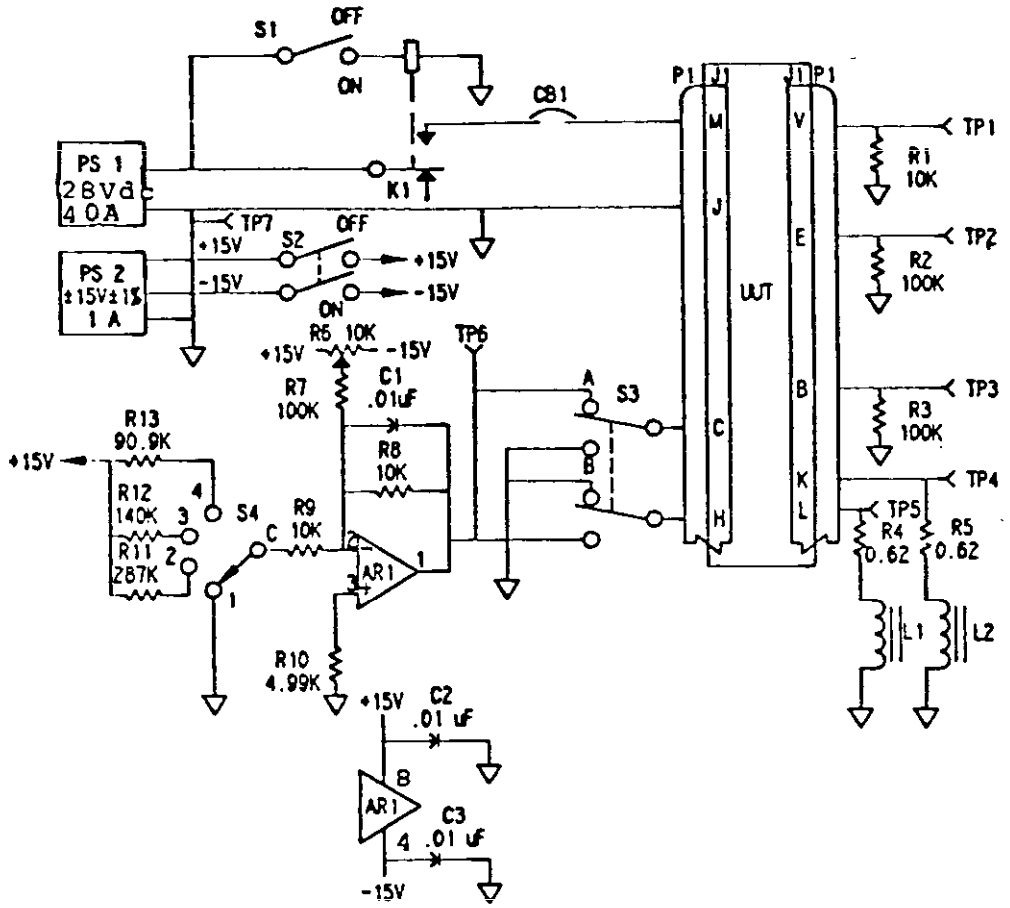
NOTE: During temperature extremes, the duty cycle at terminal L with S3 to position A, shall be 35 +/-20 for switch position S4-2 and 65 +/-20 for switch position S4-3: The duty cycle at terminal K, with S3 to position B, shall be 35 +/-20 for switch position S4-2 and 65 +/-20 for switch positon S4-3.

NOTE: Duty cycle refers to the ratio of time-on to the period expressed in percent.

NOTE: Subassemblies A1, A2 and A3 shall be subjected to their respective tests before the final assembly is subjected to the above test.



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| | | | |
|---|---|---|---|
| R1 | 10 Kohms | C2 | .1 uF |
| R2 | 100 Kohms | C3 | .1 uF |
| R3 | 100 Kohms | S1 | SPST |
| R4 | 0.62, 1.5 KW | S2 | DPDT |
| R5 | 0.62, 1.5 KW | S3 | DPDT |
| R6 | 10 Kohms, 20-turn trimmer | S4 | SP4T |
| R7 | 100 Kohms | AR1 | 12000117-3 or equivalent |
| R8 | 10 Kohms | | |
| R9 | 10 Kohms | K1 | Relay 40Adc, 24 Vdc SPST |
| Vdc | | | |
| R10 | 4.99 Kohms | CB1 | 50 A |
| R11 | 287 Kohms | L1 | 0.40 mh ± 5%, 40 A |
| R12 | 140 Kohms | L2 | 0.40 mh ± 5%, 40 A |
| R13 | 90.9 Kohms | P1 | MS3456W28-115 |
| C1 | .01 uF | | |

Note: All resistors ± 1% 1/4 watt unless otherwise specified.
NOTE: All capacitors 50 V ceramic ± 10%.

FIGURE 3. Servo amplifier test setup.



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4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS9361156. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Three servo amplifiers shall be used on each of the M163A2 and M167A2 weapon systems. One shall be used for elevation to drive the gun up or down. Two shall be used similarly for azimuth positioning of the gun.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Applicable stock number.
c. First article sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certification of conformance for each lot or shipment of product.
g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.



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6.5 Subject term (Key word) listing.

Fire Control System
Gun, anti-aircraft artillery
Vulcan Air Defense System (VADS)
Self-propelled 20MM M163A2
Towed 20MM M167A2

6.6 Changes from previous issue. The previous issues were for ARDEC internal use only. The were not forwarded to Naval Publication and Forms Center (NPFC) for publication. This issue is initial public release for unlimited distribution.

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| | MIL-S-70587B (AR) | 27 July 1991 |

3. DOCUMENT TITLE

SERVO AMPLIFIER: 9361156

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

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